

Data Pipeline Documentation

Technical workflow from raw CSV to Power BI

1. End-to-End Flow

Raw CSV → Python inspection → cleaning → rule-based enrichment → MySQL → SQL analytical views → Power BI → dashboard.

2. Pipeline Stages

Stage	Technology	Output
Data inspection	Python / Pandas	Dataset profile and quality findings.
Cleaning	Python / Pandas	cleaned_tickets.csv
Classification	Python rules	Priority, urgency, risk indicators.
Database	MySQL	Structured ticket model.
SQL analysis	MySQL SQL	Reusable analytical views and queries.
Visualization	Power BI / DAX	Three-page dashboard.

3. Final Run

Metric	Result
Input records	47,837
Output records	47,837
Failures	0
Processing time	~1 minute 47 seconds
Classification method	rule_only

4. Key Scripts

- src/data_cleaning.py
- src/rule_classifier.py
- src/run_rules_only_enrichment.py
- src/test_mysql_connection.py